Class XII Session 2025-26
Subject - Physics
Sample Question Paper - 10

Time Allowed: 30 Minutes

General Instructions:

1. There are 33 questions in all. All questions are compulsory.
2. This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
3. Use of calculators is not allowed.

[1]

BHAYAT

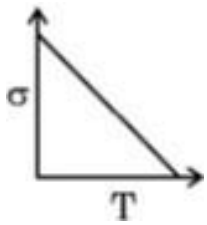
Section A

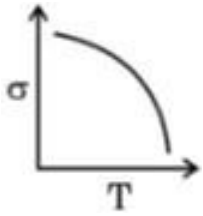
1. If N is Avogadro number, M is atomic weight and ρ, the density, the number of free electrons per unit volume of a monovalent metallic crystal can be expressed as

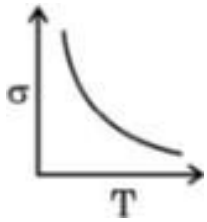
   a) $\frac{N}{\mu M}$ b) $\frac{\rho N}{M}$

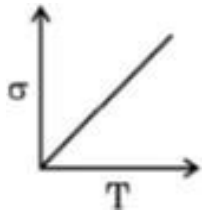
   c) $\frac{MN}{\rho}$ d) $\frac{\rho M}{N}$

2. Which one of the following is the correct representation of variation of conductivity of a conductor with temperature? **[1]**

   a) 

   b) 

   c) 

   d) 

3. The principal behind optical fibre is: **[1]**

a) Total internal reflection

b) Both Total external reflection and Total internal reflection

c) Diffraction

d) Total external reflection

4. A bar magnet of pole strength (m) and magnetic moment (M) is cut perpendicular to its axis in two equal halves. The new pole strength (m') and magnetic moment (M') of each part are respectively: **[1]**

a) m and $\frac{M}{2}$

b) m and M

c) 2m and $\frac{M}{2}$

d) $\frac{m}{2}$ and 2M

5. Two charges $3 \times 10^{-8} C$ and $-2 \times 10^{-8} C$ are located 15 cm apart. At what point on the line joining them, the potential will be zero. **[1]**

a) 3 cm from first charge

b) 9 cm from first charge

c) None of these

d) 6 cm from first charge

6. The Magnetic Resonance Imaging (MRI) is based on the phenomenon of **[1]**

a) Nuclear magnetic resonance

b) Electron paramagnetic resonance

c) Electron spin resonance

d) Diamagnetism of human tissues

7. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current: **[1]**

a) increases in both A and B.

b) increases in A and decreases in B.

c) decreases in both A and B.

d) remains same in both A and B.

8. At a point on the right bisector of a magnetic dipole, the magnetic: **[1]**

a) field is perpendicular to the axis of dipole

b) potential is zero at all points on the right bisector

c) potential varies as $\frac{1}{r^2}$

d) field varies as r^3

9. If two sources have a randomly varying phase different ϕ(t), the resultant intensity will be given by **[1]**

a) $2 I_0$

b) $\frac{3}{2} I_0$

c) $\frac{I_0}{2}$

d) $\frac{I_0}{\sqrt{2}}$

10. The number of electric field lines crossing an area ΔS is n_1 when $\Delta \vec{S} \parallel \vec{E}$, while number of field lines crossing same area is n_2 when $\Delta \vec{S}$ makes an angle of $30°$ with $\vec{E}$, then: **[1]**

a) $n_1 < n_2$

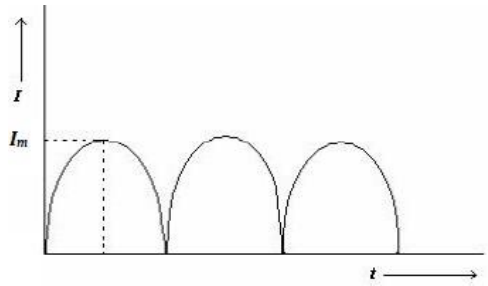
b) $n_1 = n_2$

c) $n_1 > n_2$

d) cannot say anything.

11. The output current I versus time (t) curve of a full wave rectifier is shown in the figure. The average value of the output current in this case is **[1]**

BHAYAT



a) I_m b) Zero

c) $\frac{2I_m}{\pi}$ d) $\frac{I_m}{\pi}$

12. Which of the following pairs of media has the least value of critical angle? **[1]**

a) Glass to air b) Glass to water

c) Diamond to water d) Diamond to air

13. **Assertion (A):** In his study of photoelectric emission, Hallwachs connected a negatively charged zinc plate to an electroscope. He found that negatively charged particles were emitted from the zinc plate under the action of visible light. **[1]**

Reason (R): An uncharged zinc plate becomes positively charged when it is irradiated by visible light.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion:** If the distance between parallel plates a capacitor is halved and dielectric constant is made three times then the capacitance becomes six times. **[1]**

Reason: (R) Capacitance of the capacitor does not depend upon the nature of the material of the plates of the capacitor.

a) Assertion and reason both are correct statements and reason is correct explanation for assertion.

b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

c) Assertion is correct statement but reason is wrong statement.

d) Assertion is wrong statement but reason is correct statement.

15. **Assertion (A):** Interference pattern is made by using yellow light instead of red light, the fringes becomes narrower. **[1]**

Reason (R): In Young's double slit experiment, fringe width is given by $\beta = \frac{\lambda D}{d}$.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. **Assertion (A):** In alternating current, the direction of motion of free electrons changes periodically. **[1]**

Reason (R): Alternating current changes its direction after a certain is always constant.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

BHAYAT